2 Figure 2.1 shows a diagram of a car brake pedal. The driver applies a force of 14 N to the end of the brake pedal. The piston applies a force of 72 N to the brake pedal.

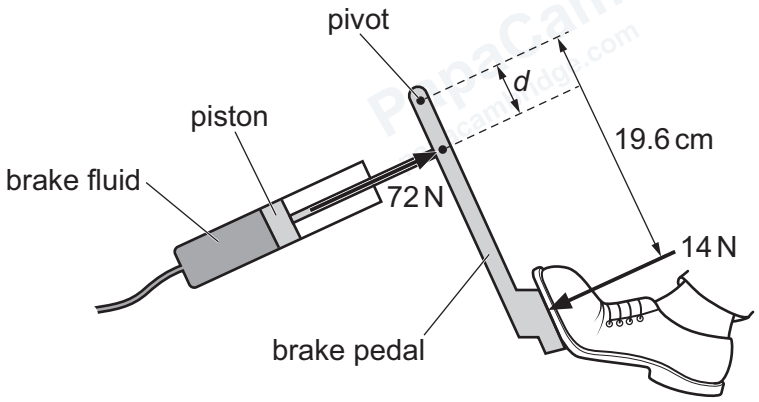

Figure 2.1

(a) Show that the moment of the 14 N force about the pivot is approximately 270 N cm.

[1]

(b) The brake pedal is stationary. The piston exerts a force of 72 N on the brake pedal.

Calculate the distance *d*.

d = ... cm [2]

(c) The pressure exerted by the brake fluid on the piston is 8.4×10^4 Pa.

Calculate the cross-sectional area of the piston.

cross-sectional area = ... m^2 [2]

[Total: 5]

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